神经网络基础

张彦博

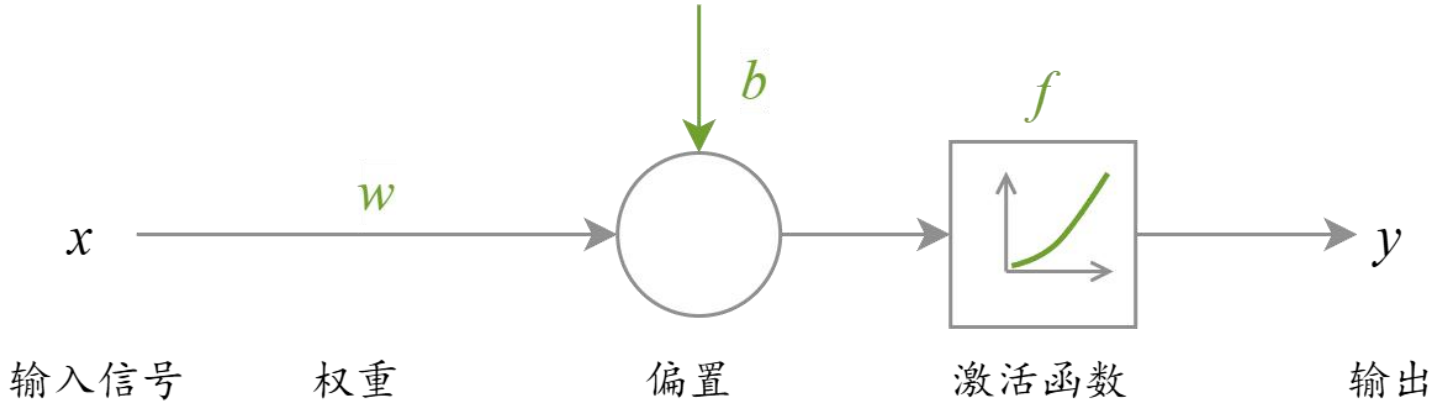
国科大杭高院智能科学与技术学院

zhangyb031204@163.com

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1.神经元

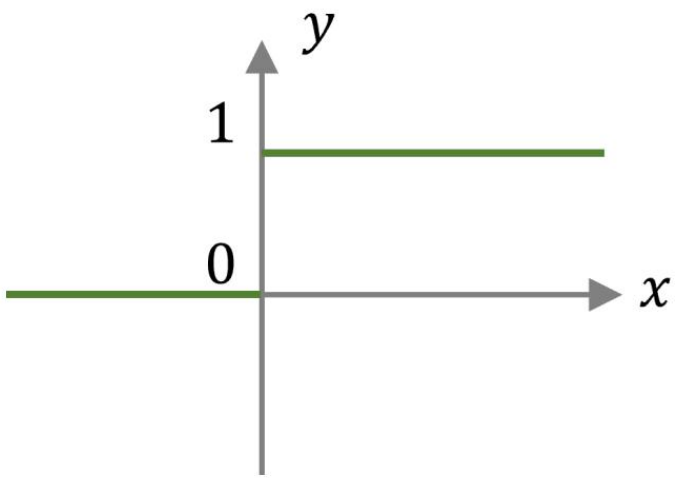


$$y = f(w \cdot x + b)$$

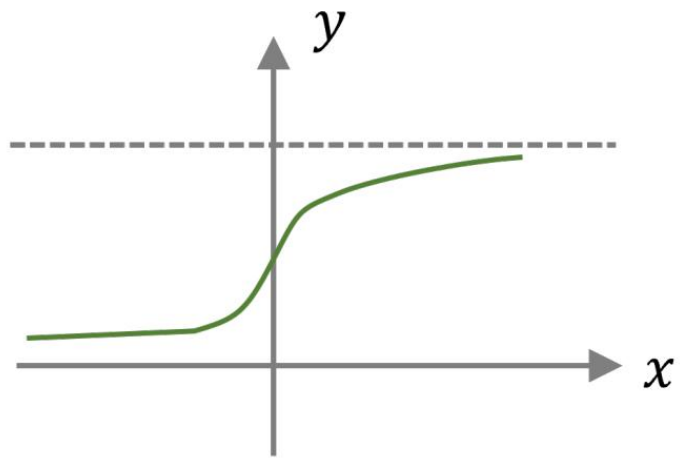
可训练参数
权重
偏置

超参数
激活函数

1. 神经元：激活函数



sgn阶跃函数



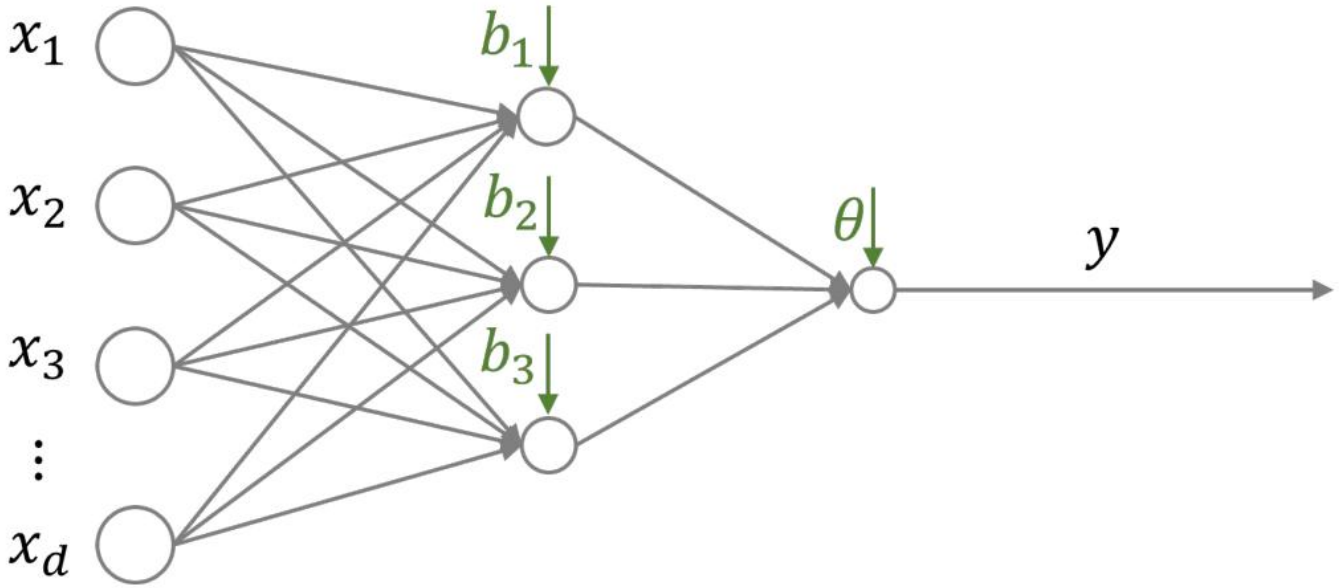
sigmoid函数

$$y(\boldsymbol{x}) = \text{sgn}\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} - \theta\right)$$

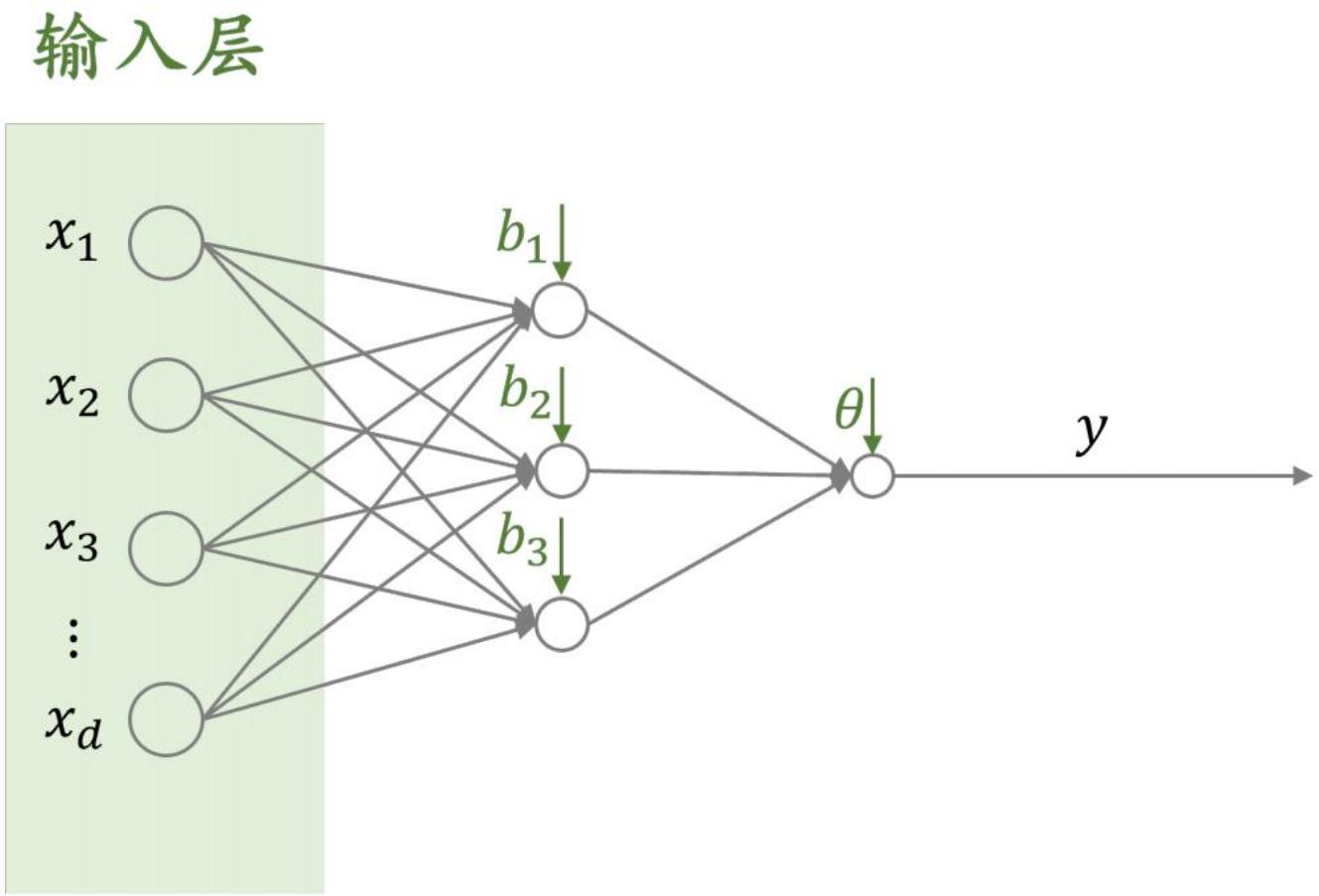
$$y(\boldsymbol{x}) = \text{sigmoid}\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} - \theta\right) = \frac{1}{1 + e^{(-\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + \theta)}}$$

激活函数是神经网络获得非线性表达能力的关键

2.神经网络：基本结构

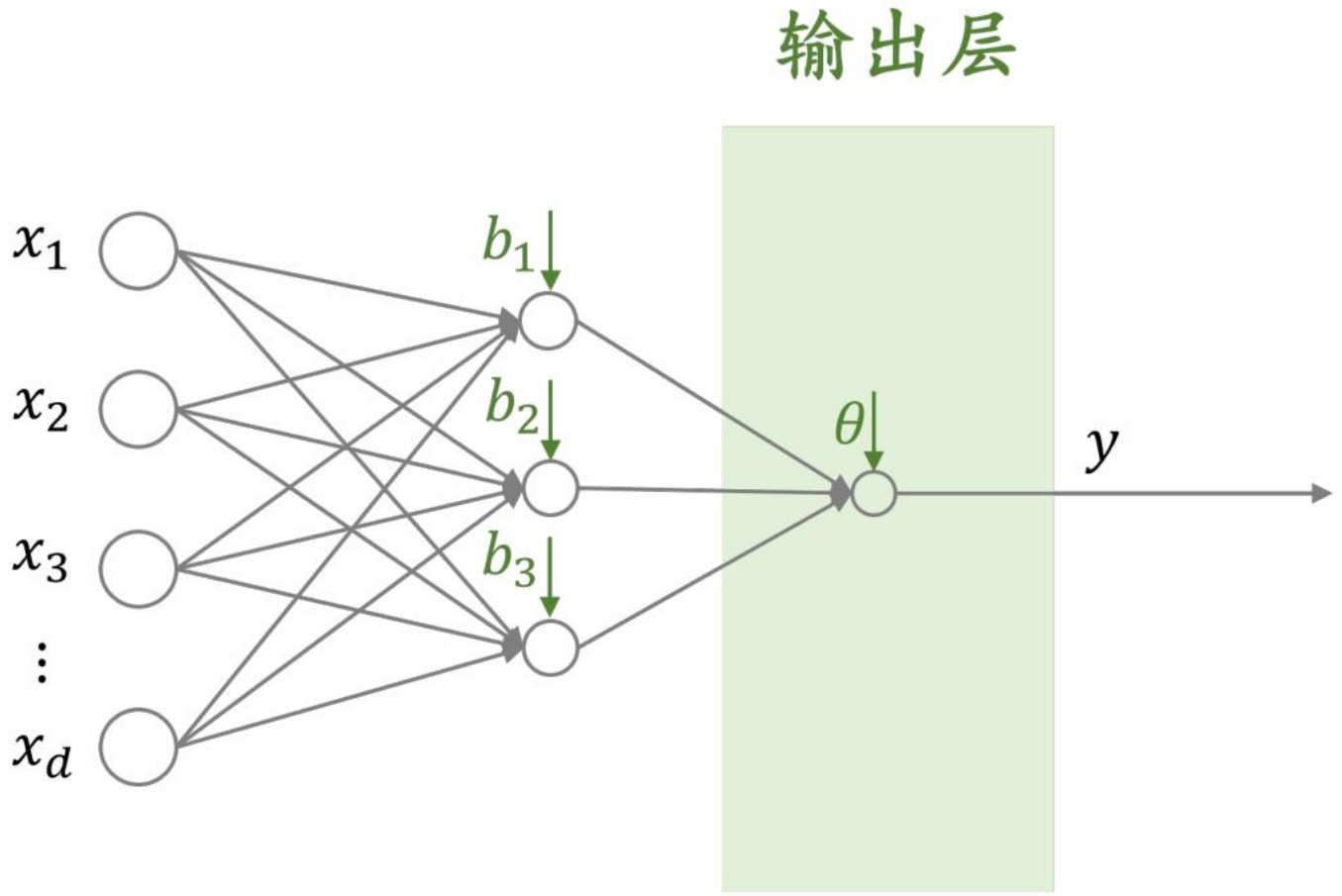


2.神经网络：基本结构



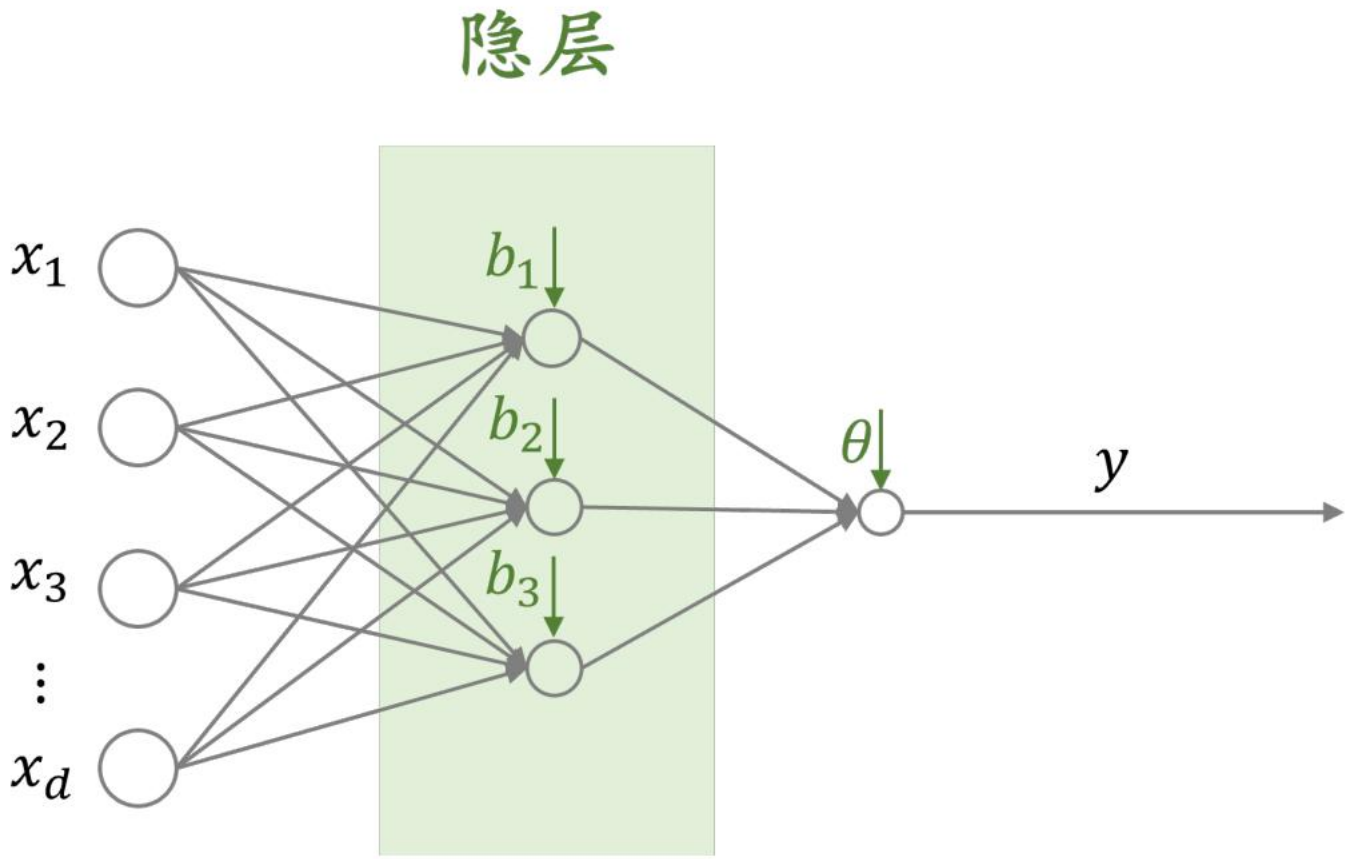
- 输入层没有神经元
- 节点数与特征数相同

2.神经网络：基本结构



- 输入层有神经元
- 回归问题1个节点，k 分类问题 k 个节点

2.神经网络：基本结构

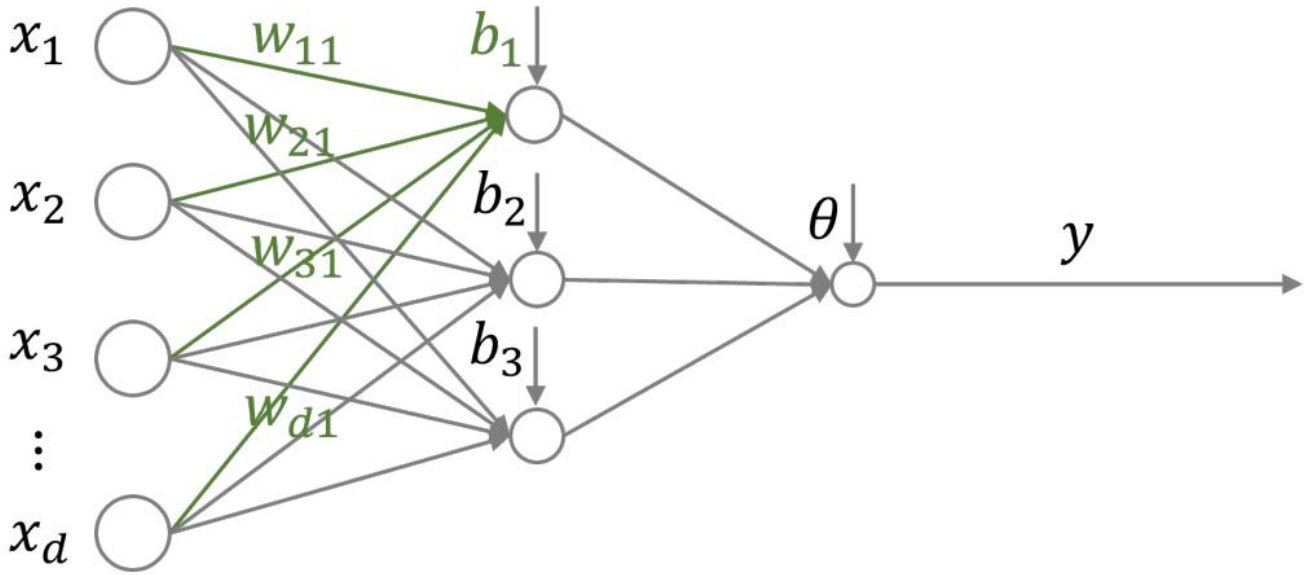


- 隐含层有神经元
- 隐含层层数，每层节点数可以看作超参数

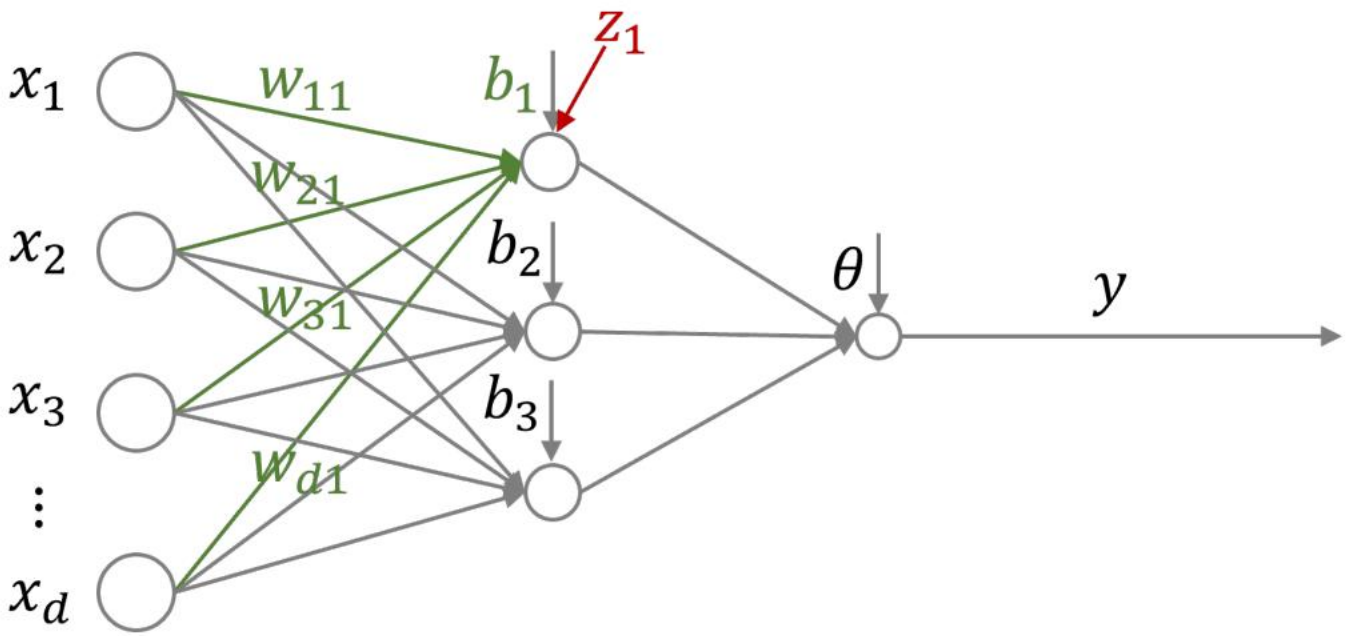
2. 神经网络：推理and训练

	推理	训练
核心定义	使用已训练好的模型，输入数据直接输出预测结果	输入标注数据，迭代调整参数，让模型预测越来越准
核心流程	前向传播 （输入→逐层计算→输出）	两步循环： ①前向传播算结果 ②反向传播算梯度、更新参数
参数状态	权重w、偏置b固定，不更新	权重w、偏置b迭代更新

2.神经网络：前向传播



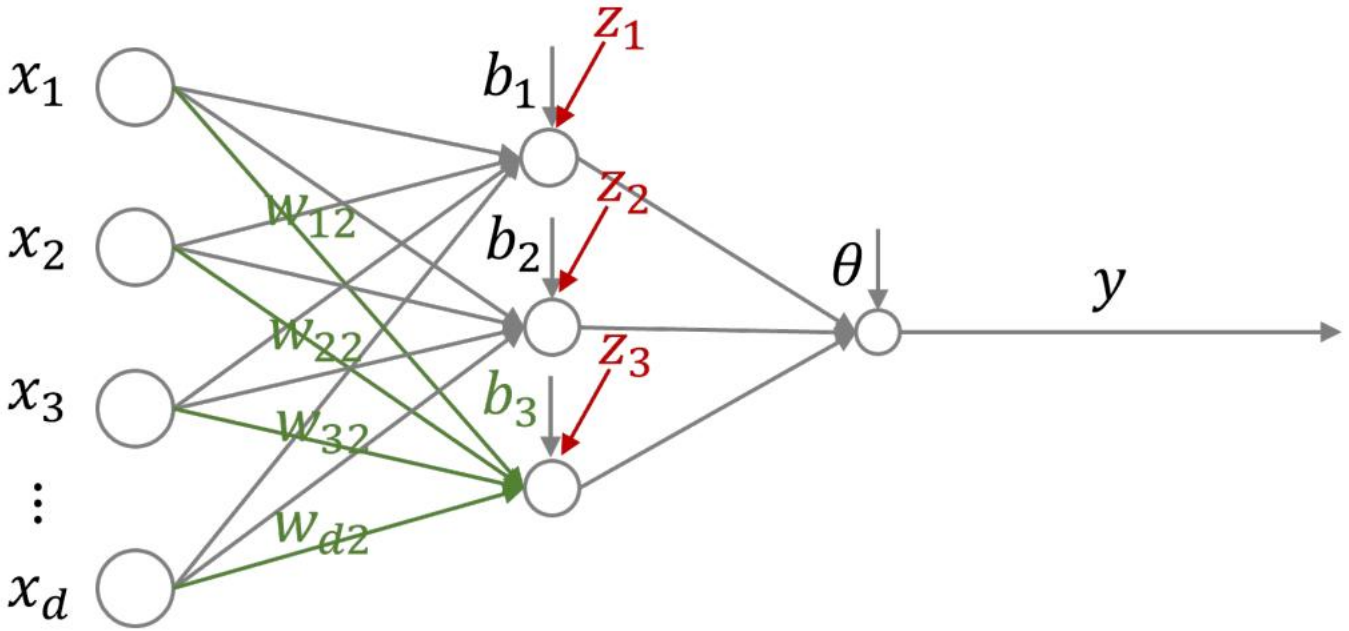
2.神经网络：前向传播



$$z_1 = w_{11}x_1 + w_{21}x_2 + \cdots + w_{d1}x_d - b_1$$

$$z_1 = \boldsymbol{w}_1\boldsymbol{x} - b_1, \quad \boldsymbol{w}_1 = [w_{11}, w_{21}, \cdots, w_{d1}]$$

2.神经网络：前向传播

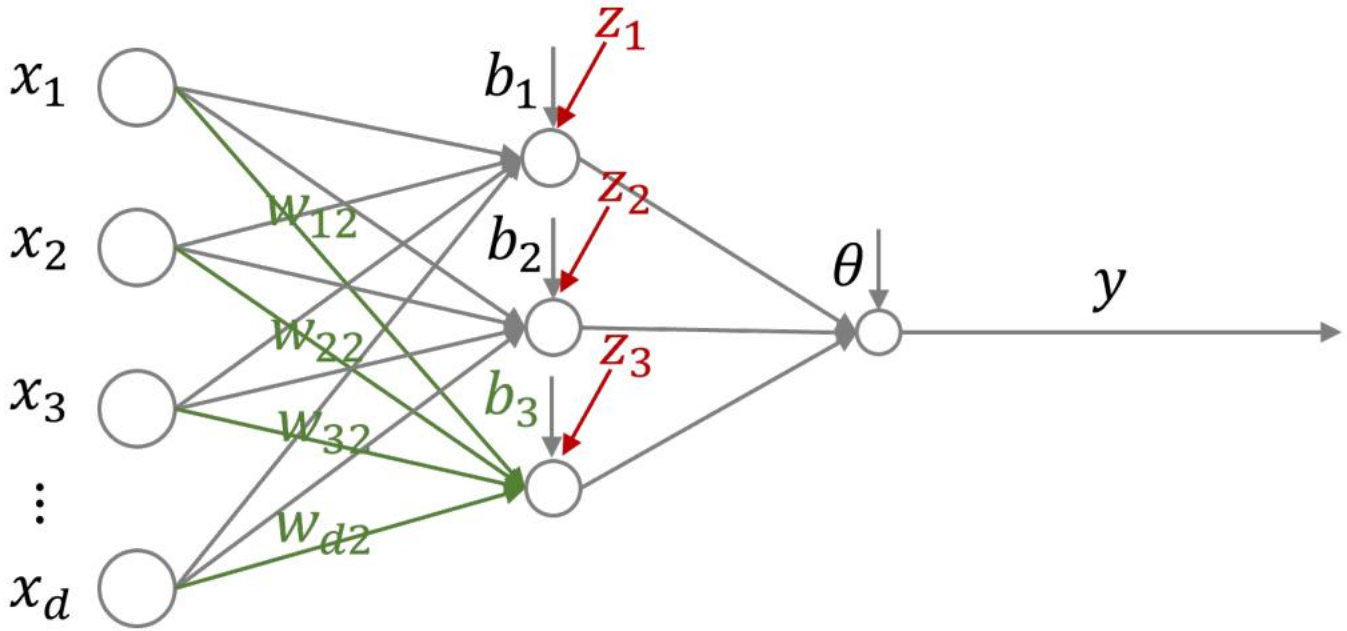


$$z_1 = \boldsymbol{w}_1\boldsymbol{x} - b_1, \quad \boldsymbol{w}_1 = [w_{11}, w_{21}, \cdots, w_{d1}]$$

$$z_2 = \boldsymbol{w}_2\boldsymbol{x} - b_2, \quad \boldsymbol{w}_2 = [w_{12}, w_{22}, \cdots, w_{d2}]$$

$$z_3 = \boldsymbol{w}_3\boldsymbol{x} - b_3, \quad \boldsymbol{w}_3 = [w_{13}, w_{23}, \cdots, w_{d3}]$$

2.神经网络：前向传播



$$z_1 = \boldsymbol{w}_1\boldsymbol{x} - b_1$$

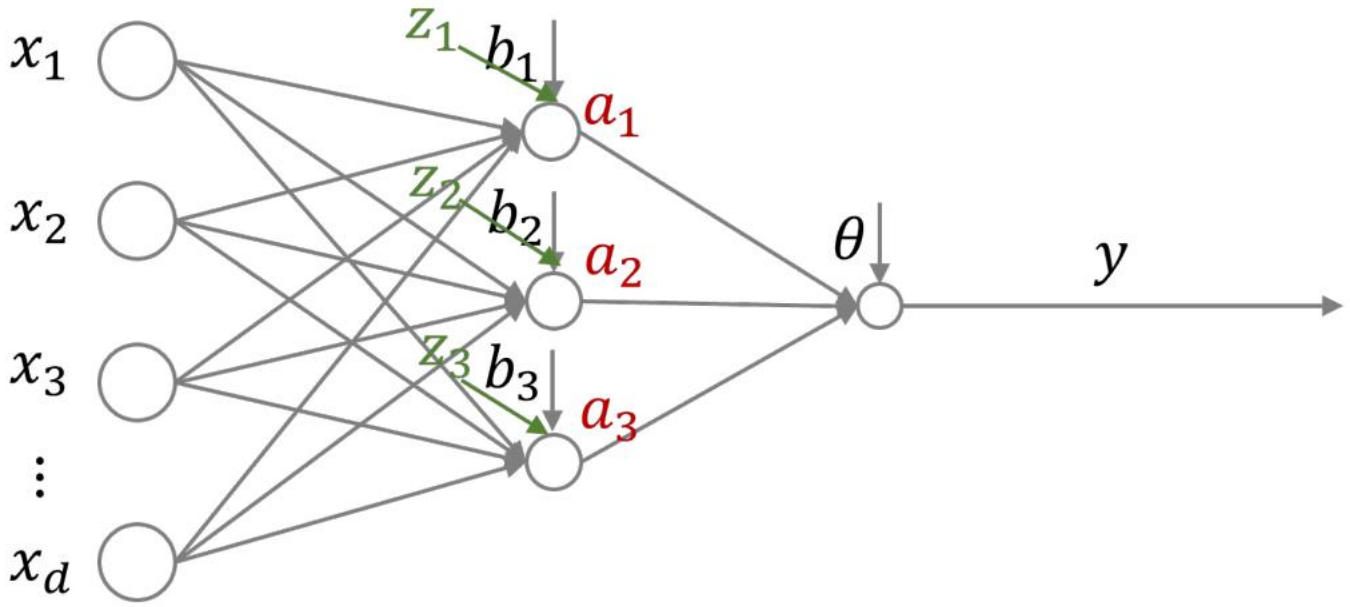
$$z_2 = \boldsymbol{w}_2\boldsymbol{x} - b_2$$

$$z_3 = \boldsymbol{w}_3\boldsymbol{x} - b_3$$

$$\boldsymbol{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \quad \boldsymbol{W} = \begin{bmatrix} w_{11} & w_{21} & \cdots & w_{d1} \\ w_{12} & w_{22} & \cdots & w_{d2} \\ w_{13} & w_{23} & \cdots & w_{d3} \end{bmatrix} \quad \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_3 \end{bmatrix} \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\boldsymbol{z} = \boldsymbol{W}\boldsymbol{x} - \boldsymbol{b}$$

2.神经网络：前向传播



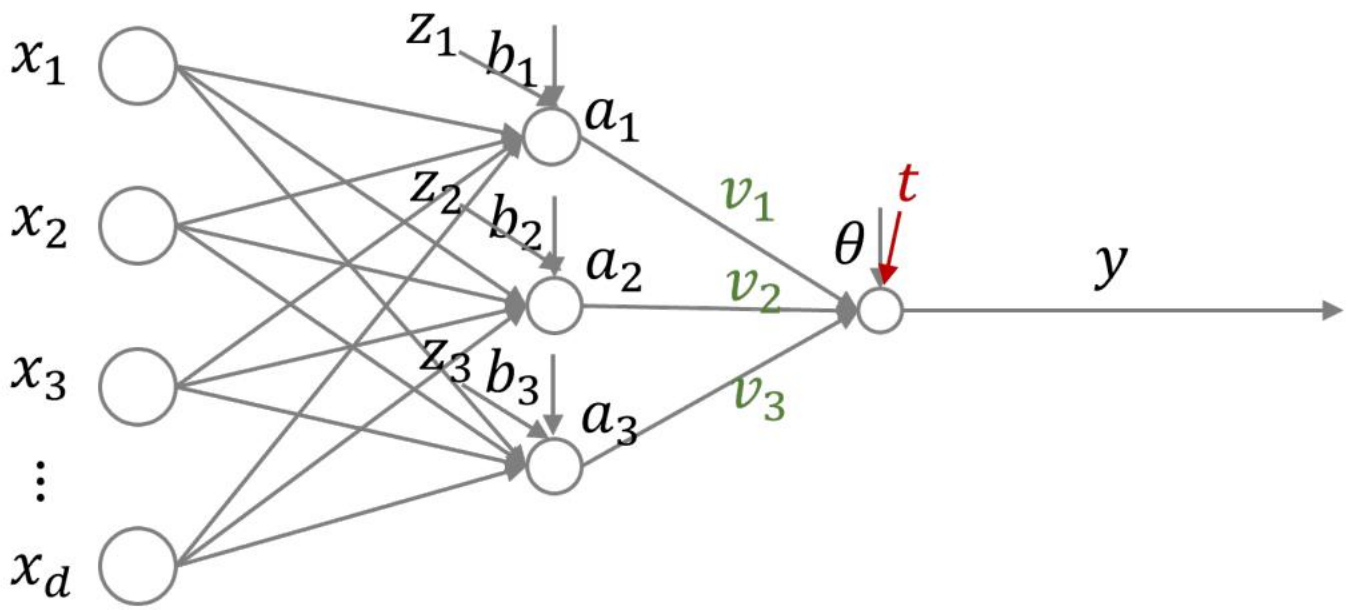
$$a_1 = \sigma(z_1)$$

$$a_2 = \sigma(z_2)$$

$$a_3 = \sigma(z_3)$$

$$\boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z})$$

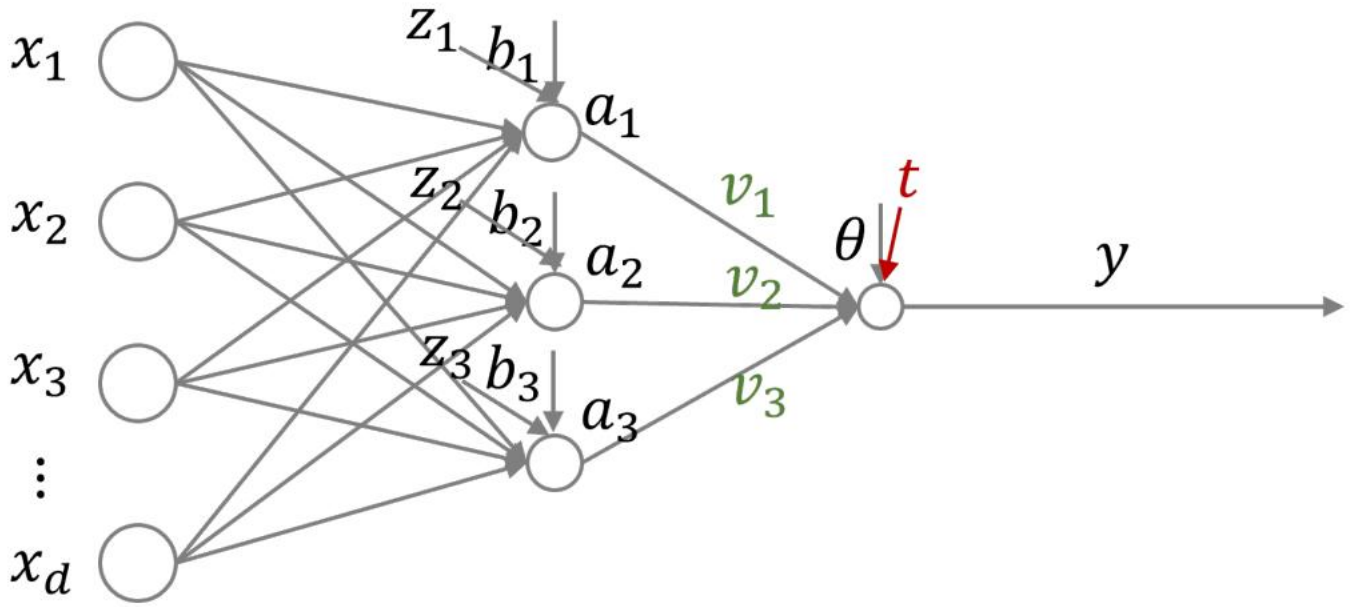
2.神经网络：前向传播



$$t = \boldsymbol{va} - \theta$$

$$y = \sigma(t)$$

2.神经网络：前向传播



$$\boldsymbol{z} = \boldsymbol{W}\boldsymbol{x} - \boldsymbol{b}$$

$$\boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z})$$

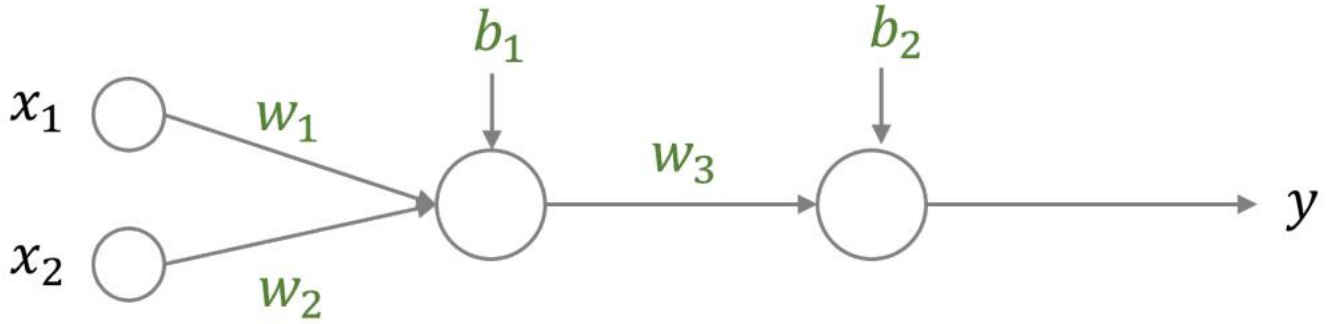
$$t = \boldsymbol{v}\boldsymbol{a} - \theta$$

$$y = \sigma(t)$$

$$y = \sigma\left(\boldsymbol{v}\boldsymbol{\sigma}\left(\boldsymbol{W}\boldsymbol{x} - \boldsymbol{b}\right) - \theta\right)$$

2.神经网络：前向传播

有如图所示的神经网络结构，网络所有节点激活函数均为 sigmoid 函数，其中 $w_1 = 2$，$w_2 = 1$，$w_3 = 4$，$b_1 = 3$，$b_2 = 5$，试计算神经网络对样本 $[1, 2]^{\mathrm{T}}$ 的预测值。



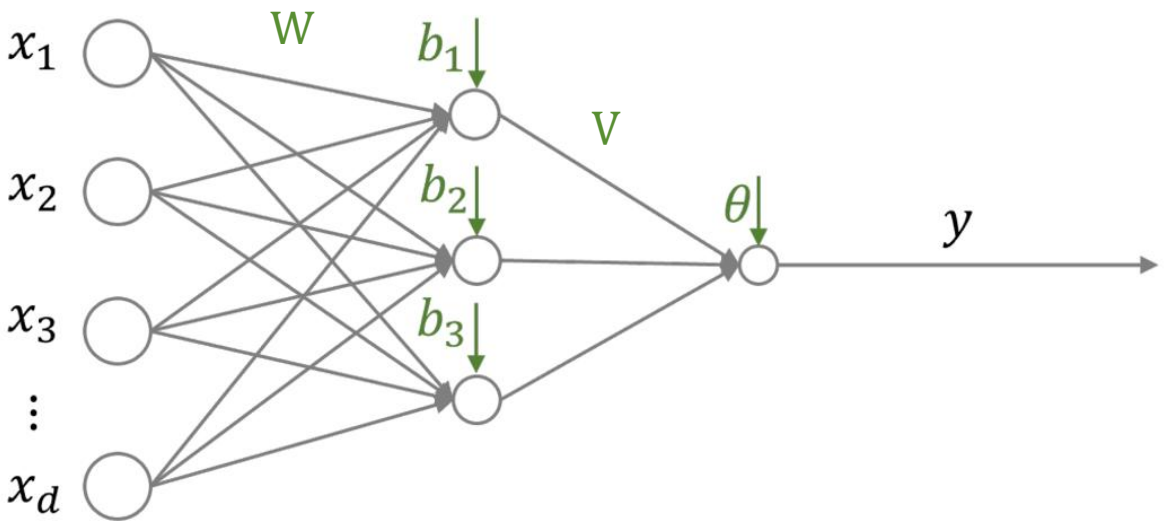
$$z = w_1 x_1 + w_2 x_2 - b_1 = 2 \times 1 + 1 \times 2 - 3 = 1$$

$$a = \sigma(z) = 0.7311$$

$$t = w_3 a - b_2 = 4 \times 0.7311 - 3 = -0.0758$$

$$y = \sigma(t) = 0.4811$$

2. 神经网络：反向传播



① 前向传播，得到预测输出y

$$y = \sigma(v\sigma(Wx - b) - \theta)$$

② 计算损失（Loss）

$$L = Loss(y, y_{true})$$

③ 反向传播，逐层计算梯度

$$\nabla_W = \frac{\partial L}{\partial W}, \nabla_b = \frac{\partial L}{\partial b}, \nabla_v = \frac{\partial L}{\partial v}, \nabla_\theta = \frac{\partial L}{\partial \theta}$$

④根据梯度更新参数

$$W = W - \eta\nabla_W$$

$$b = b - \eta\nabla_b$$

$$v = v - \eta\nabla_v$$

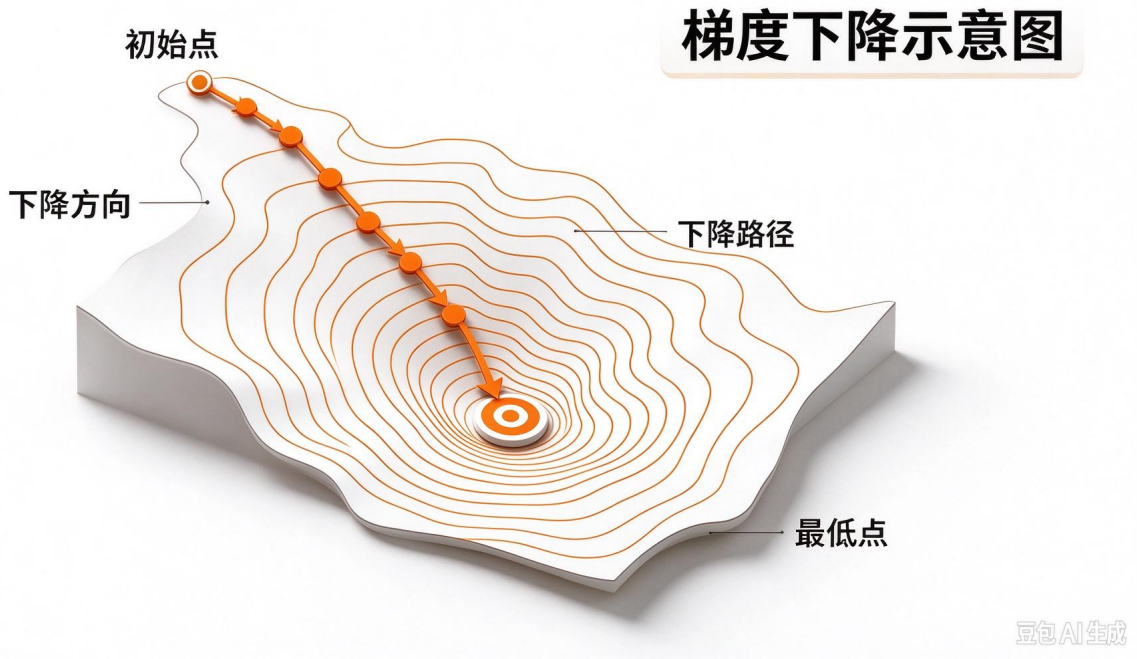
$$\theta = \theta - \eta\nabla_\theta$$

损失：量化预测和真实值之间的差距，训练目标就是最小化L
这里选用的是**均方误差**

η：学习率

分布式训练中，每张卡只算自己的梯度，需要全局梯度同步 → 产生大量AllReduce通信

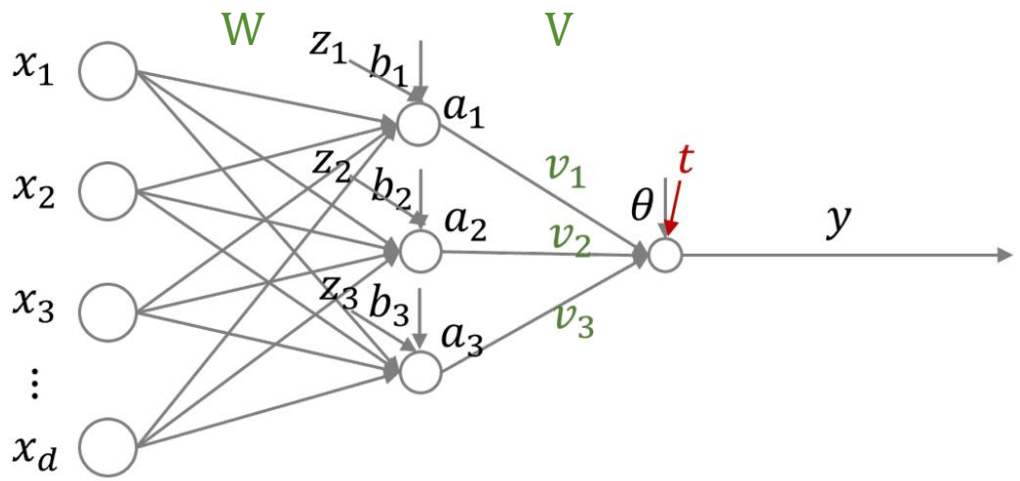
2. 神经网络：反向传播



梯度：是一个向量，方向是函数在该点上升最快的方向

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n}\right)$$

2. 神经网络：反向传播



$$z = Wx - b$$

$$a = \sigma(z)$$

$$t = va - \theta$$

$$y = \sigma(t)$$

$$y = \sigma(v\sigma(Wx - b) - \theta)$$

$$L = \frac{1}{2}(y - y_{true})^2$$

$$\nabla_W = \frac{\partial L}{\partial W}, \nabla_b = \frac{\partial L}{\partial b}, \nabla_v = \frac{\partial L}{\partial v}, \nabla_\theta = \frac{\partial L}{\partial \theta}$$

$$W = W - \eta\nabla_W$$

$$b = b - \eta\nabla_b$$

$$v = v - \eta\nabla_v$$

$$\theta = \theta - \eta\nabla_\theta$$

$$\frac{\partial L}{\partial \theta} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial \theta} = (y - y_{true}) \circ y \circ (1 - y)$$

$$\frac{\partial L}{\partial v} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial v} = (y - y_{true}) \circ y \circ (1 - y)\, a^{\mathrm{T}}$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a}\frac{\partial a}{\partial z}\frac{\partial z}{\partial b} = v^{\mathrm{T}}[(y - y_{true}) \circ y \circ (1 - y)] \circ a \circ (1 - a)$$

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a}\frac{\partial a}{\partial z}\frac{\partial z}{\partial W} = v^{\mathrm{T}}[(y - y_{true}) \circ y \circ (1 - y)] \circ a \circ (1 - a)\, x^{\mathrm{T}}$$

重点内容

- 神经元结构：输入信号、权重、偏置、激活函数、输出
- 超参数
- 神经网络模型结构
- 推理/训练
- 前向传播
- 反向传播
    - 梯度下降